

Demonstrate Kubernetes Dashboard and CLI for Cluster Management

Step 1 - Initialize minikube

- Run *minikube start* and *minikube status* to start and check the status of minikube

```
(~/Desktop/devops/lab9) (1rv24mc038_shreerakshahr@asus:pts/0)
(09:30:32) -> minikube start
minikube v1.37.0 on Debian bookworm/sid
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Restarting existing docker container for "minikube" ...
! Failing to connect to https://registry.k8s.io/ from inside the minikube container
💡 To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
  ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
  ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, default-storageclass, dashboard
🏡 Done! kubectrl is now configured to use "minikube" cluster and "default" namespace by default
(~/Desktop/devops/lab9) (1rv24mc038_shreerakshahr@asus:pts/0)
(09:30:59) -> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 2 - Write service.yaml

```
(~/Desktop/devops/lab10) (1rv24mc038_shreerakshahr@asus:pts/1)
(09:46:26) -> cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: NodePort
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
```

Step 3 - Write deployment.yaml

```
(~/Desktop/devops/lab10) (1rv24mc038_shreerakshahr@asus:pts/1)
(09:46:11) -> cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Step 4 - Apply YAML files

```
(~/Desktop/devops/lab10) (1rv24mc038_shreerakshahr@asus:pts/1)
(09:46:41) -> kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created

(~/Desktop/devops/lab10) (1rv24mc038_shreerakshahr@asus:pts/1)
(09:47:24) -> kubectl apply -f service.yaml
service/nginx-service created
```

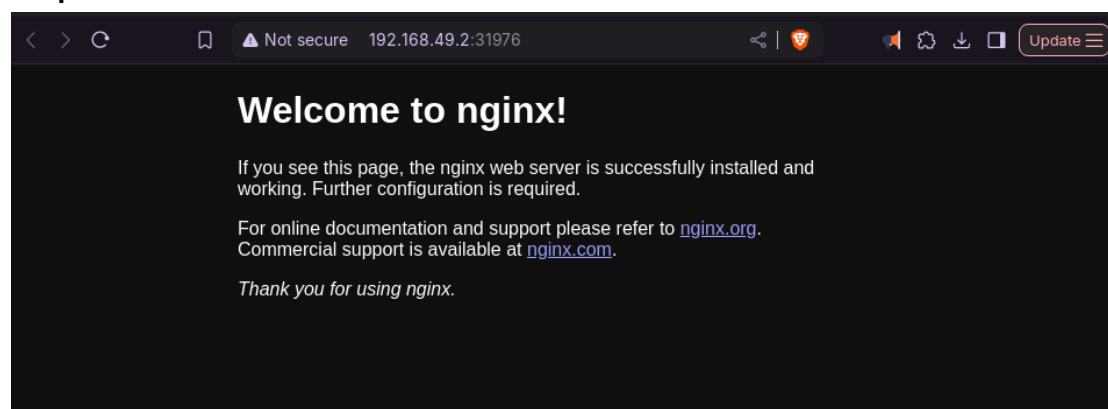
Step 5 - Start service in minikube

```
(~/Desktop/devops/lab10) (1rv24mc038_shreerakshahr@asus:pts/1)
(09:47:27) -> minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:31976

🚀 Opening service default/nginx-service in default browser...

Output -



Step 6 - View minikube Dashboard

```
(~/Desktop/devops/lab10) (1rv24mc038_shreerakshahr@asus:pts/2)
(09:48:17) -> minikube dashboard (Fri, Dec 19)

🤖 Verifying dashboard health ...
🚀 Launching proxy ...
🤖 Verifying proxy health ...
🌐 Opening http://127.0.0.1:43639/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
🔑 Opening in existing browser session.
```

Output -

The screenshot displays the Kubernetes Dashboard interface. The top navigation bar shows the 'Workloads' section is active. The left sidebar lists various Kubernetes resources. The main content area is divided into three sections: 'Workload Status', 'Deployments', and 'Pods'.

Workload Status: This section provides a high-level overview of the cluster's health. It features three large green circles, each representing a different workload type: Deployments (Running: 1), Pods (Running: 2), and Replica Sets (Running: 1).

Deployments: This section shows a table of active deployments. The table has columns for Name, Images, Labels, Pods, and Created. A single deployment, 'nginx-deployment', is listed with the image 'nginx:latest' and label 'app: nginx'. It shows 2/2 pods running and was created 37 minutes ago.

Pods: This section shows a detailed table of all pods in the cluster. The table includes columns for Name, Images, Labels, Node, Status, Restarts, CPU Usage, Memory Usage, and Created. Two pods are listed, both running on the 'minikube' node. The first pod is 'nginx-deployment-6f9664446b-12k67' and the second is 'nginx-deployment-6f9664446b-172sh'. Both have a status of 'Running' and 0 restarts.

Name	Images	Labels	Pods	Created
nginx-deployment	nginx:latest	app: nginx	2 / 2	37 minutes ago

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)	Created
nginx-deployment-6f9664446b-12k67	nginx:latest	app: nginx pod-template-hash: 6f9664446b	minikube	Running	0	-	-	37 minutes ago
nginx-deployment-6f9664446b-172sh	nginx:latest	app: nginx pod-template-hash: 6f9664446b	minikube	Running	0	-	-	37 minutes ago